Ziyun Liu

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RESEARCH OUTLINE

My research focuses on mathematically principled optimization and the training dynamics of large foundation models. I develop adaptive optimizers for low-rank fine-tuning using matrix geometry and curvature [3,4]. Beyond algorithm design, I study how data geometry shapes optimization and model behavior. My work on Muon establishes conditions under which data-induced curvature makes spectral normalization advantageous [2], while my MoE research investigates the near-additivity of expert contributions within a layer through empirical analysis and a tractable training model [1].

I am broadly interested in developing principled ways of understanding and improving large foundation models.

EDUCATION

The Hong Kong University of Science and Technology — **Hong Kong SAR**
Ph.D. in Mathematics — Sep. 2024 – Jun. 2028 (expected)
- Advisor: Prof. Jian-Feng Cai; GPA: 3.95
- Research on optimization and its applications in the training of deep neural networks.

M.Phil. in Mathematics — Sep. 2022 – Aug. 2024
- Advisor: Prof. Jian-Feng Cai; GPA: 3.99
- Thesis: Greedy Algorithms for Matrix Approximation via the Interlacing Polynomial Method

Tongji University — **Shanghai, China**
B.Eng. in Computer Science (Minor in Mathematics) — Sep. 2018 – July 2022
- Advisor: Prof. Jae Kyu Choi and Prof. Hanli Wang
- Thesis: Improved Sparse Signal Restoration Algorithm Based on Data-Driven Tight Frames

EXPERIENCE

LongCat Pretraining Team, Meituan — **Shenzhen, China**
Research Intern — Apr. 2026 – Present
- Working with Dr. Jiaqi Zhang and Chao Wang on principled methods for pretraining large foundation models.

Sea AI Lab — **Singapore (Remote)**
Associate Member — Oct. 2025 – Apr. 2026
- Worked with Dr. Chao Du and Dr. Tianyu Pang on manifold-constrained optimization for LLMs and MoE training dynamics.

Imaging Team, SmartMore Inc. — **Shenzhen, China**
Research Intern — July 2021 – May 2022
- Advised by Dr. Sixing Hu, I worked on computational imaging, including video denoising, white balancing, and color constancy.

Department of Mathematics, Tongji University — **Shanghai, China**
Research Assistant — May 2020 – May 2021
- Advised by Prof. Jae Kyu Choi, I worked on compressed sensing for spectral sparse signal restoration and NMR imaging.

PUBLICATIONS (*equal contribution)

[1] On the Additivity of Expert Contributions in Mixture-of-Experts Models — Under Review at ICLR 2027
Ziyun Liu, Fengzhuo Zhang, Chao Du, Tianyu Pang, Jian-Feng Cai — [PDF]

[2] Provable Advantages of Muon from Data-Induced Curvature — Under Review at ICLR 2027
Jian-Feng Cai, **Ziyun Liu**, Juntao You, Fengzhuo Zhang — [PDF]

[3] AdaPreLoRA: Adafactor Preconditioned Low-Rank Adaptation — **NeurIPS 2026**
Ziyun Liu, Fengmiao Bian, Jian-Feng Cai — [PDF]

[4] Finding Low-Rank Matrix Weights in DNNs via Riemannian Optimization: RAdaGrad and RAdamW — **NeurIPS 2025**
Fengmiao Bian, Jinyang Zheng*, **Ziyun Liu***, Jianzhou Luo, Jian-Feng Cai — [PDF]

ACADEMIC SERVICE

- Reviewer for ICML 2026, NeurIPS 2026, ICLR 2026 & 2027, and AAAI 2027

AWARDS

- Top Reviewer, NeurIPS 2026 — Oct. 2026
- Silver Reviewer, ICML 2026 — May 2026
- Din-Yu Hsieh Teaching Award (Best TA Award), HKUST — May 2024
- Outstanding Undergraduate Student Scholarship, Tongji University — Nov. 2020, Nov. 2021

TEACHING

- MATH 2111 Matrix Algebra and Applications — Fall 2023, Fall 2024, Summer 2025, Fall 2025, Fall 2026
- MATH 2023 Multivariable Calculus — Spring 2024
- MATH 1014 Calculus II, MATH 1003 Calculus and Linear Algebra — Spring 2023, Spring 2026

SKILLS

Languages: Python, C/C++, MATLAB, Chinese (Native), English (Fluent) **Platform and Toolkits:** Bash, Git, Vim, Slurm, PyTorch